

COMPOSITE MATERIAL DATASHEETS CFRP IMA/M21E-UD

DEFENCE AND SPACE

Part:1.4
Chapter: 1491
Revision: 1.5
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MATERIAL IDENTIFICATION

EU 1E001 - This data is controlled under EU Dual Use Regulation 2021/821, as amended. Authorization may be required for the export of the data outside of the EU.

Short informal name	IMA/M21E-UD
Commercial name	Hexcel Hexply M21E/34%/UD194/IMA-12K
Specifications	Airbus: ABS5377 B (AIMS05-27-002 B) CASA: Z-19.791
Description	High Performance Carbon Fibre Reinforced Epoxy Prepreg Unidirectional Tape / 180°C Curing Class / Intermediate Modulus
Version	1.5 (*) Add DVs for 90°C/Wet conditions

REFERENCES

- [1] TEA-T-NT-180036 Is. C "LAMINATES STRENGTH. DISCUSSION ON TESTS RESULTS FOR IMA/M21E: PLAIN STRENGTH AND RESIDUAL STRENGTH AFTER IMPACT."
- [2] QTRSM332 Is. 3 "Qualification of M21E/IMA UD Tape. QTR - Qualification Test Report_AP2039_AP2295."
- [3] TE-T-NT-230038 Is. A "DERIVATION OF MATERIAL DESIGN VALUES FOR CFRP UD-TAPE PRE-PREG HEXCEL IMA/M21E"

PHYSICAL PROPERTIES

Cured Ply Thickness (CPT)	0.184 mm
Density	1600 kg/m ³
Tg-onset (wet)	130 °C
Max. Operational Limit (MOL)	>100 °C

Coefficients of Thermal Expansion CTE (1/°C)	Range	-50...0°C	0...50°C	50...150°C
	α_1	-1.0E-07	-4.7E-07	-5.6E-07
	α_2	2.9E-05	2.9E-05	2.9E-05
	α_3	2.9E-05	2.9E-05	2.9E-05

ELASTIC MODULI

Property / Units	E_1 (MPa)	E_2 (MPa)	G_{12} (MPa)	ν_{12}	K_B
Values	154000	8500	4200	0.35	0.85

Property / Units	E_3 (MPa)	G_{13} (MPa)	G_{23} (MPa)	ν_{13}	ν_{23}
Values	8500	4200	2500	0.35	0.5

Remarks:

- Consolidated design values are common to Airbus Commercial
- Min. test K_B ratio for the (10/80/10) laminate was $K_B=0.828$. Selected design value is $K_B=0.85$, valid from QI data and more representative
- Estimated ν_{23} truncated

PLAIN STRENGTHS

Ultimate stresses of the ply (from UD-laminates testing).

Property	Mean Values (MPa)				B-basis K_B	E-KDF			B-basis Values (MPa)			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
F_{1t}	3303	3303	2996	-	0.9	1	0.907	-	2973	2973	2696	-
F_{1c}	1557	1327	1189	-	0.830	0.852	0.764	-	1292	1101	987	-
F_{2t}	56.2	36	31.2	-	0.789	0.641	0.555	-	44.3	28.4	24.6	-
F_{2c}	288	189	161	-	0.900	0.656	0.559	-	259.2	170.1	144.9	-
F_{12}	115.9	88.7	75.5	59.8	0.900	0.765	0.651	0.516	104.3	79.8	68.0	53.8

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Design values for checking the laminate strength using the modified Maximum Strain Criterion.

Property	Mean Values (µε)				B-basis K_B	E-KDF			B-basis Values (µε)			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
ϵ_{tail}	17300	17300	15500	-	0.896	1	0.896	-	15500	15500	13795	-
ϵ_{call}	10100	8600	7600	-	0.830	0.851	0.757	-	8400	7100	6300	-

INTERLAMINAR PLY STRENGTHS

Property	Mean Values				B-basis K_B	E-KDF			B-basis Values			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
F_{13} (MPa)	106.5	69.6	61.5	52.5	0.900	0.654	0.577	0.493	95.9	62.6	55.4	47.2
F_{3t} (MPa)	F_{3t} shall be made equal to F_{2t} in QI and directed laminates if thickness ≥ 3 mm, otherwise contact Stress Methods dpt.											
F_{3c} (MPa)	Through-thickness compression strength F_{3c} shall be made equal to F_{2c}											
G_{IC} (N/mm)	0.300	-	-	-	0.8	-	-	-	-	-	-	-
G_{IIC} (N/mm)	0.800	-	-	-	0.8	-	-	-	-	-	-	-

OPEN HOLE AND BOLTED JOINTS

Reference material strengths in open-hole, filled-hole and bearing for the quasi-isotropic laminate.

Property	Mean Values (MPa)				B-basis K_B	E-KDF			B-basis Values (MPa)			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
OHT_{QI}	555.8	555.8	551.6	-	0.900	1	0.993	-	500.2	500.2	496.5	-
OHC_{QI}	304.9	251.8	212.0	-	0.900	0.826	0.695	-	274.5	226.6	190.8	-
FHT_{QI}	496.8	496.8	482.0	-	0.900	1	0.970	-	447.1	447.1	433.8	-
FHC_{QI}	391.5	313.8	269.0	-	0.827	0.802	0.687	-	323.8	259.5	223.0	-
F_{BR-QI}	1064	814	748.0	-	0.863	0.765	0.703	-	917.8	702.1	648.5	-

COMPOSITE DAMAGE TOLERANCE

- Residual strength after transverse impact

The following TAI, CAI model is valid for laminates following the standard design rules (symmetric, balanced, with minimum 10% of plies at angles 0°, 90°, 45° and -45°).

Property	B-basis Value for all conditions
TAI (µε)	8000

$$E/t^2 = A \cdot (1 - e^{-B \cdot d(t)})$$

Approximation model type: CDTM-T04

$$CAI = K_B \cdot EKDF \cdot \{ CAI_{\infty-QI} / (E_{\theta} / E_{QI})^{C_x} + A / e^{B_e \cdot (E/t^2)} \}$$

Property	Design Value
A (J/mm ²)	6.8
B	4.5

Property	Mean Value RT/Dry	B-basis K_B	E-KDF		
			70°C/W	90°C/W	120°C/W
$CAI_{\infty-QI}$ (µε)	3560	0.83	0.87	0.70	-
A (µε)	3500				
B_e	0.6				
C_x	0.81				

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...where:

- E is the impact energy, not higher than the cut-off identified during damage threats analysis (for many parts of the airframe, 35J as cut-off value shall be acceptable)
- t the laminate thickness
- d_0 the dent depth after impact (typ. 2.0-2.5mm for BVID detectability under a GVI inspection)
- E_{QI} is the elastic modulus of the quasi-isotropic laminate (57593 MPa in this case)
- E_θ is the modulus of the laminate object of analysis in the direction of interest

Another approximation model is proposed next.

Approximation model type: CDTM-T07

$$CAI = K_B \cdot EKDF \cdot \{ CAI_{\infty-QI} \cdot (3 / K_{TOH})^C \cdot [1 + A / e^{B \cdot (E/t^\alpha)}] \}$$

Property	Mean Value RT/Dry	B-basis KB	E-KDF		
			70°C/W	90°C/W	120°C/W
$CAI_{\infty-QI}$ (µε)	3300	0.83	0.86	0.70	-
α	2.00				
A	1.09				
B	0.73				
C	1.02				

...where:

- K_{TOH} is the stress concentration factor in the laminate due to an open circular hole
- A, B, C and exponent α are parameters of the model to be adjusted from tests data.

(*) CAI shall be calculated not only for the 0° direction (X parallel to material 0° reference), but also for the rest of ply orientations, i.e. the 45° and 90° directions, by rotating the X axis and re-evaluating previous expressions with the new values of E_x , E_y , G_{xy} and ν_{xy} . In a general case CAI strength shall be expressed as 3 scalar values with the allowable strains applicable to plies at 0° (CAI_0), at $\pm 45^\circ$ (CAI_{45}) and at 90° (CAI_{90}).

- Residual strength after transverse BVID near lightening holes

No design data for CAI_{hole} still available

- Residual strength after edge impact

No design data for CAI_E still available